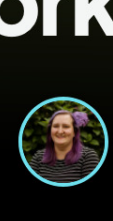
Article

7 Types of AI Agents to Automate Your Workflows in 2025



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The conversation around AI has shifted from chatbots—those basic interfaces designed to respond to user queries—to more sophisticated AI agents. AI agents are autonomous programs that can observe their environment, make decisions, and take actions to achieve specific goals. They can monitor data streams, automate complex workflows, and execute tasks without constant human supervision.

As businesses seek more sophisticated automation solutions, these agents are growing in popularity; 79% of organizations have already adopted AI agents, and 66% of those organizations have already seen measured productivity, according to a recent survey from PwC.

This growth is fueled by advances in natural language processing (NLP), a push for more personalized customer experiences, and increased automation for more repetitive tasks.

Read on to learn more about the types of AI agents and use cases to better understand the application of these intelligent agents.

Key takeaways:

- AI agents are systems that use AI to complete tasks on your behalf. They use machine learning and natural language processing.
- They can complete simple tasks for specific programs with minimal human interaction in areas such as finance, healthcare, customer service, retail, manufacturing, and industry.
- There are seven types of AI agents: simple reflex agents, model-based reflex agents, goal-based agents, learning agents, utility-based agents, hierarchical agents, and multi-agent systems.

What is an AI agent?

AI agents are systems that use AI to complete tasks without the need for human prompts or intervention. They rely on machine learning and natural language processing to analyze an environment, collect information about how certain changes would affect it, problem-solve, and determine what actions to take to complete specific tasks.

AI agents have two main components for their architecture: their software and the agent program that contains the decision-making algorithms. They also incorporate several modules that help it collect data, consider the possibilities, and make decisions. These are the profiling, memory, planning, and action modules.

The level of autonomy and decision-making capabilities that AI agents have set them apart from virtual assistants and chatbots, which mainly rely on scripts, user requests, and conversational AI to function. AI agents can assess the environment's data, draw conclusions from that data, and perform actions completely unprompted.

Benefits of AI agents

Having the ability to deploy agents that can make decisions, evaluate data, and respond to specific requests can assist with workflow automation, increase personal productivity, and help organizations outsource specific tasks. The main benefits of AI agents include:

- **Increased efficiency:** Organizations can use AI agents to automate repetitive tasks such as claims processing, appointment scheduling, or customer inquiries.
- **Improved accuracy:** AI agents can analyze patterns and make data-driven decisions, which results in more accurate decisions for tasks that require extensive data analysis or pattern detection.
- **Personalization:** In using input data for tasks such as shopping preferences or healthcare treatment plans, AI agents can take specifications and create a personalized experience that accounts for individual factors or preferences. These could be suggested products for online shopping based on your past purchases.
- **Learning and adaptability:** With their models and training, AI agents can adapt over time and integrate new feedback to create more updated guidelines. An example would include streaming service recommendations.

Challenges of AI agents

Even with all the benefits of AI agents, it's still important to consider some of the complexities that come along with AI deployment and maintenance. For agentic AI, they include:

- **Computational costs and resources:** Having the available technology to run AI agents and systems can require significant computing power, storage, and memory resources, as well as trained staff. This requires sizable upfront costs and extensive planning for proper deployments.
- **Human training and oversight:** Though AI agents can act autonomously, they do require some human training and general oversight to ensure the models are operating properly. This requires a lot of available data and trained professionals who know how to train, calibrate, and update AI models.
- **Integration difficulties:** Not all AI agent types can work together in hybrid or mult-agent systems. This requires testing how different agents are compatible before deployment to avoid costly mistakes or interoperability errors.
- **Infinite loops:** This occurs when the AI agent starts an action chain that leads back to the original action and starts an endless cycle. This can affect data quality and use up costly resources.

There are also agent-specific challenges, such as being able to adjust to dynamic environments, requiring accurate data, needing planning algorithms, depending on utility functions to effectively operate, and risks of overfitting data.

How do AI agents work?

AI agents range from simple task-specific programs to sophisticated systems that combine perception, reasoning, and action capabilities. The most advanced agents operate through a cycle of processing inputs, make decisions, and execute actions while continuously updating their knowledge.

Perception and input processing

AI agents begin by gathering and processing input from their environment. This could include parsing text commands, analyzing data streams, or receiving sensor data. The perception module converts raw inputs into a format the agent can understand and process. For example, when a customer submits a support request, an AI agent could process the ticket by analyzing text content, user history, and metadata like priority level and timestamp.

Decision-making and planning

Using machine learning models like NLP, sentiment analysis, and classification algorithms, agents evaluate their inputs against their objectives. These models work together: NLP first processes and understands the input text, sentiment analysis evaluates its tone and intent, and classification algorithms determine which category of response is most appropriate. This layered approach enables agents to process complex inputs and respond appropriately. They generate possible actions, assess potential outcomes, and select the most appropriate response based on their programming and current context. For instance, when handling the support ticket, the AI agent could evaluate content and urgency to determine whether to handle it directly or escalate to a human agent.

Knowledge management

Agents maintain and use knowledge bases that contain domain-specific information, learned patterns, and operational rules. Through Retrieval-Augmented Generation (RAG), agents can dynamically access and incorporate relevant information from their knowledge base when forming responses. In our support ticket example, the agent uses RAG to pull information from product documentation, past cases, and company policies to generate accurate, contextual solutions rather than relying solely on its training data.

Action execution

Once a decision is made, agents execute actions through their output interfaces. This could involve generating text responses, updating databases, triggering workflows, or sending commands to other systems. The action module ensures the chosen response is properly formatted and delivered. Continuing our example, the customer support agent might then send automated troubleshooting steps, route the ticket to a specialized department, or flag it for immediate human attention.

Learning and adaptation

Advanced AI agents can improve their performance over time through feedback loops and learning mechanisms. They analyze the outcomes of their actions, update their knowledge bases, and refine their decision-making processes based on success metrics and user feedback. Using reinforcement learning techniques, these agents develop optimal policies by balancing exploration (trying new approaches) with exploitation (using proven successful strategies). In the support scenario, the agent learns from resolution success rates and satisfaction scores to improve its future responses and routing decisions, treating each interaction as a learning opportunity to refine its decision-making model.

Types of AI agents

Businesses have a rich but complex landscape of AI agent options, ranging from simple task-specific automation tools to sophisticated multi-purpose assistants that can transform entire workflows. The choice or development of an AI agent depends on several factors—including technical complexity, implementation costs, and specific use cases—with some organizations opting for ready-to-use solutions while others invest in custom agents tailored to their unique needs.

1. Simple reflex agents

Simple reflex agents are one of the most basic forms of artificial intelligence. These agents make decisions based solely on their current sensory input, responding immediately to environmental stimuli without needing memory or learning processes. Their behavior is governed by predefined condition-action rules, which specify how to react to particular inputs.

Though they are limited in complexity, this straightforward approach makes them efficient and easy to implement, especially in environments where the range of possible actions is limited.

Key components:

- **Sensors**: Much like human senses, these gather information from the environment. For a simple reflex agent, sensors are typically basic input devices that detect specific environmental conditions like temperature, light, or motion.
- **Condition-action rules**: These predefined rules determine how the agent responds to specific inputs. The logic is direct—if the agent detects a specific condition, it immediately performs a corresponding action.
- **Actuators**: These execute the decisions made by the agent, translating them into physical or digital responses that alter the environment in some way, such as activating a heating system or turning on lights.

Use cases: Simple reflex agents are ideal for transparent, predictable environments with limited variables.

- **Industrial safety sensors** that immediately shut down machinery when detecting an obstruction in the work area.
- **Automated sprinkler systems** that activate based on smoke detection.
- **Email auto-responders** that send predefined messages based on specific keywords or sender addresses.
- **Safety management:** Intenseye provides real-time safety management and serious injury and fatality detection that connects with closed-circuit television systems to get visual data of your workspace and collect insights. You can then provide data to the AI system about which areas are designed for specific tasks and what type of actions to look for within the warehouse or manufacturing facility. It can then provide alerts and preventative safety measure recommendations.
- **Fraud detection:** Feedzai offers a platform to help detect financial crime and fraud. It can be used by retail banks, commercial banks, business owners, and others to scan financial transaction data and identify any potential suspicious activity, transaction fraud, and money laundering activity, and notify the necessary individuals via alert.

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2. Model-based reflex agents

Model-based reflex agents are a more advanced form of intelligent agents designed to operate in partially observable environments. Unlike simple reflex agents, which react solely based on current sensory input, model-based agents maintain an internal representation, or model, of the world.

This model tracks how the environment evolves, allowing the agent to infer unobserved aspects of the current state. While these agents don't actually "remember" past states in the way more advanced agents do, they use their world model to make better decisions about the current state.

Key components:

- **State tracker**: Maintains information about the current state of the environment based on the world model and sensor history.
- **World model**: This model contains two key types of knowledge: how the environment evolves independent of the agent and how the agent's actions affect the environment.
- **Reasoning component**: Uses the world model and current state to determine appropriate actions based on condition-action rules.

Use cases: These agents are suitable for environments where the current state isn't fully observable from sensor data alone.

- **Smart home security systems**: Using models of normal household activity patterns to distinguish between routine events and potential security threats.
- **Quality control systems**: Monitoring manufacturing processes by maintaining a model of normal operations to detect deviations.
- **Network monitoring:** Selector uses AIOps to provide AI-driven network monitoring and issue discovery. The AI model relies on metrics, logs, events, and network metadata to learn about overall network conditions and regular operating standards. It will then use the AI model to detect anomalies, route alerts, and help with root cause analysis.
- **Self-driving cars:** Waymo makes self-driving cars, which rely on sensors, cameras, data, and AI systems to navigate through cities and roads. The car's perception system uses sensors and computer vision to respond to pedestrians, traffic signals, and other road obstacles. It uses detailed maps and real-time sensor data to provide the AI system with information about the car's location and surroundings.

3. Goal-based agents

Goal-based agents are designed to pursue specific objectives by considering the future consequences of their actions. Unlike reflex agents that act based on rules or world models, goal-based agents plan sequences of actions to achieve desired outcomes. They use search and planning algorithms to find action sequences that lead to their goals.

Key components:

- **Goal state**: A clear description of what the agent aims to achieve
- **Planning mechanism**: The ability to search through possible sequences of actions that could lead to the goal.
- **State evaluation**: Methods to assess whether potential future states move closer to or further from the goal.
- **Action selection**: The process of choosing actions based on their predicted contribution toward reaching the goal.
- **World model**: Understanding of how actions change the environment, used for planning.

Use cases: Goal-based agents are suited for tasks with clear, well-defined objectives and predictable action outcomes.

- **Smart heating systems**: Planning temperature adjustments to reach desired comfort levels efficiently.
- **Inventory management systems**: Planning reorder schedules to maintain target stock levels.
- **Project management systems:** Productivity tools, such as ClickUp, help you organize tasks, set new goals, and gain insights into how you spend your time. Its ClickUp Brain AI analyzes project data and information so you can ask questions about project status, create new goals for tasks, and summarize project updates. It also has features to automate certain tasks, such as meeting scheduling, notetaking, project brief creation, and task creation and assignment based on team and project data.
- **Robotic warehouse systems:** Symbotic offers an end-to-end AI robotic warehouse automation platform that integrates autonomous robots. These robots use vision and sensing technologies to move inventory, adapt to handle different product types, and recognize and build out paths throughout the warehouse. This is paired with the AI software, which can increase overall robotic control, get insights about performance, and efficiently optimize pallets on the floor.

4. Learning agents

A learning agent is an artificial intelligence system capable of improving its behavior over time by interacting with its environment and learning from its experiences. These agents modify their behavior based on feedback and experience, using various learning mechanisms to optimize their performance. Unlike simpler agent types, they can discover how to achieve their goals through experience rather than purely relying on pre-programmed knowledge.

Key components:

- **Performance element**: The component that selects external actions, similar to the decision-making modules in simpler agents.
- **Critic**: Provides feedback on the agent's performance by evaluating outcomes against standards, often using a reward or performance metric.
- **Learning element**: Uses critic's feedback to improve the performance element, determining how to modify behavior to do better in the future.
- **Problem generator**: Suggests exploratory actions that might lead to new experiences and better future decisions.

Use cases: Learning agents are suited for environments where optimal behavior isn't known in advance and must be learned through experience.

- **Industrial process control**: Learning optimal settings for manufacturing processes through trial and error.
- **Energy management systems**: Learning patterns of usage to optimize resource consumption.
- **Quality control systems**: Learning to identify defects more accurately over time.
- **Streaming services:** Netflix relies on machine learning and learning agents to provide content recommendations, learn your preferences for content, and provide a more personalized content experience. These agents also learn how you interact with content, navigate the Netflix catalog, and integrate any feedback you provide for show or movie recommendations.
- **Customer service chatbots:** Ada offers an AI customer service agent that provides information to users, regardless of location or language. It can provide insights from conversations, help test out new responses and answer options, get feedback from both users and developers, and provide answers via chat, voice, or email.

5. Utility-based agents

A utility-based agent makes decisions by evaluating the potential outcomes of its actions and choosing the one that maximizes overall utility. Unlike goal-based agents that aim for specific states, utility-based agents can handle tradeoffs between competing goals by assigning numerical values to different outcomes.

Key components:

- **Utility function**: A mathematical function that maps states to numerical values, representing the desirability of each state.
- **State evaluation**: Methods to assess current and potential future states in terms of their utility.
- **Decision mechanism**: Processes for selecting actions that are expected to maximize utility.
- **Environment model**: Understanding of how actions affect the environment and resulting utilities.

Use cases: Utility-based agents are suited for scenarios requiring a balance between multiple competing objectives.

- **Resource allocation systems**: Balancing machine usage, energy consumption, and production goals.
- **Scheduling systems**: Balancing task priorities, deadlines, and resource constraints.
- **Stock trading bots:** Intellectia has a financial AI agent, which can provide tailored financial advice based on market data, trends, and your investing preferences. It is driven by real-time custom data from the market, analyst opinions, market sentiment, technical indicators, and charts to generate responses. It also offers an AI stock picker to provide stock picks before the market opens.
- **Smart building management:** Siemens' Building X uses AI to help address the challenges of building management and build better smart buildings that can proactively respond to changes in climate, security incidents, infrastructure conditions, and sustainability levels. Through unified management and AI algorithms, you can get a centralized view of everything that is happening within your buildings and process the data to gain greater operational insight.

6. Hierarchical agents

Hierarchical agents are structured in a tiered system, where higher-level agents manage and direct the actions of lower-level agents. This architecture breaks down complex tasks into manageable subtasks, allowing for more organized control and decision-making.

Key components:

- **Task decomposition**: Breaks down complex tasks into simpler subtasks that can be managed by lower-level agents.
- **Command hierarchy**: Defines how control and information flow between different levels of agents.
- **Coordination mechanisms**: Ensures that different levels of agents work together coherently.
- **Goal delegation**: Translates high-level objectives into specific tasks for lower-level agents.

Use cases: Hierarchical agents are best suited for systems with clear task hierarchies and well-defined subtasks.

- **Manufacturing control systems:** Coordinating different stages of production processes.
- **Building automation:** Managing basic systems like HVAC and lighting through layered control.
- **Smart factories:** Applied Materials' SmartFactory uses AI to help with quality control, signal management, standards compliance, dynamic scheduling, and latency reduction. It can also help with equipment maintenance, scheduling, and supply chain optimization. Being able to use an AI system that can improve and coordinate different stages of manufacturing and provide insights can lead to less downtime, increased overall production, and more informed decision-making.
- **Robotics:** Boston Dynamics' Atlas uses AI and robotics to create a hierarchical system that can help with planning, movement, and decision-making. With the integration of sensors, cameras, and a base level of available actions, it can use the AI system to figure out what task to do, how to do it, and then send signals to robotic motors to actually complete the tasks. Atlas can then use reinforcement learning to get feedback on task loops or action chains.

7. Multi-agent system (MAS)

A multi-agent system involves multiple autonomous agents interacting within a shared environment, working independently or cooperatively to achieve individual or collective goals. While often confused with more advanced AI systems, traditional MAS focuses on relatively simple agents interacting through basic protocols and rules.

Types of multi-agent systems:

- **Cooperative systems**: Agents share information and resources to achieve common goals. For example, multiple robots working together on basic assembly tasks.
- **Competitive systems**: Agents compete for resources following defined rules. Like multiple bidding agents in a simple auction system.
- **Mixed systems**: Combines both cooperative and competitive behaviors, such as agents sharing some information while competing for limited resources.

Key components:

- **Communication protocols**: Define how agents exchange information.
- **Interaction rules**: Specify how agents can interact and what actions are permitted.
- **Resource management**: Methods for handling shared resources between agents.
- **Coordination mechanisms**: Systems for organizing agent activities and preventing conflicts.

Use cases: MAS is best suited for scenarios with clear interaction rules and relatively simple agent behaviors.

- **Warehouse management**: Multiple robots coordinating to move and sort items.
- **Basic manufacturing**: Coordinating simple assembly tasks between multiple machines.
- **Resource allocation**: Managing shared resources like processing time or storage space.
- **AI-based research:** Anthropic recently built its own MAS research system to help provide more in-depth research responses with its Claude models. Using multi-agent systems for research purposes means that you can search through more information much more quickly through multiple subagents, and receive an answer that is more accurate than a single agent doing sequential searches across the internet.
- **Traffic management and optimization:** Miovision offers Miovision Adaptive, a software that helps improve traffic signal performance and reduce overall traffic congestion. It uses multiple, decentralized AI agents to collect traffic data, provide optimization feedback, integrate vehicle and pedestrian traffic flows, and collect connected vehicle data.

Resources

- RAG, AI Agents, and Agentic RAG: An In-Depth Review and Comparative Analysis
- Build Real-Time AI Agents with GenAI and Serverless Functions
- Building Local AI Agents: A Guide to LangGraph, AI Agents, and Ollama

AI agents FAQs

What are AI agents and how do they differ from chatbots?

AI agents are autonomous programs that can observe their environment, make decisions, and take actions to achieve specific goals without constant human supervision, while chatbots are basic interfaces designed primarily to respond to user queries. AI agents can monitor data streams, automate complex workflows, and execute tasks independently, representing a more sophisticated evolution beyond simple conversational interfaces.

How do AI agents work and make decisions?

AI agents work through a cycle of perception (gathering data from their environment), decision-making (processing information and determining appropriate actions), and execution (implementing chosen responses through output interfaces). Advanced agents can improve over time through feedback loops and reinforcement learning, analyzing outcomes and refining their decision-making processes based on success metrics.

What are the main functions of an AI agent?

An AI agent's primary functions include perceiving an environment's dynamic conditions, acting to change those conditions, interpreting its own perceptions of the environment, problem-solving, and determining actions.

How many types of AI agents are available?

There are seven types of AI agents: simple reflex agents, model-based reflex agents, goal-based agents, learning agents, utility-based agents, hierarchical agents, and multi-agent systems.

What are the primary applications of AI agents?

AI agents can be used in customer support, fraud detection, basic IT troubleshooting, marketing, finance, healthcare, manufacturing, predictive analytics, and supply chain optimization. Of course, new applications for AI agents are appearing on a regular basis.

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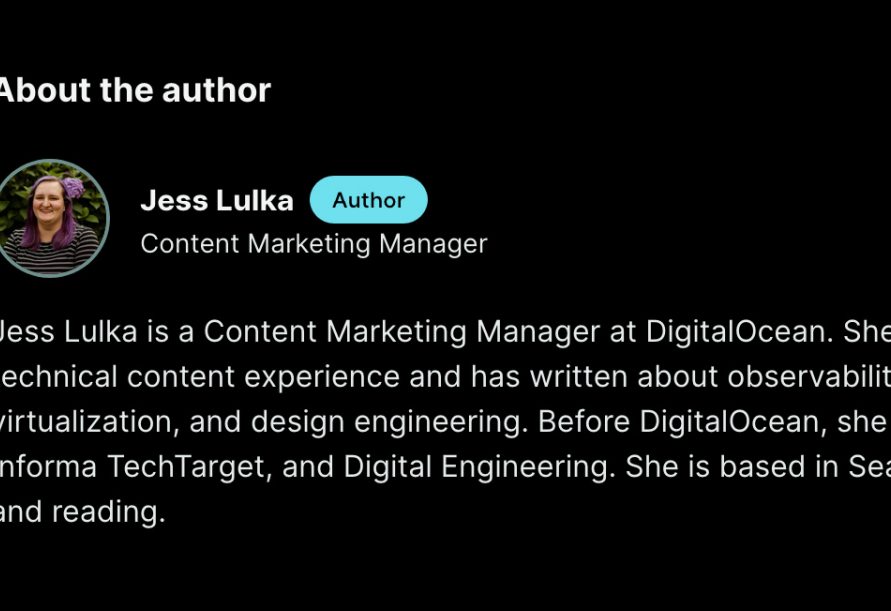

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Jess Lulka is a Content Marketing Manager at DigitalOcean. She has over 10 years of B2B technical content experience and has written about observability, data centers, IoT, server virtualization, and design engineering. Before DigitalOcean, she worked at Chronosphere, Informa TechTarget, and Digital Engineering. She is based in Seattle and enjoys pub trivia, travel, and reading.

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